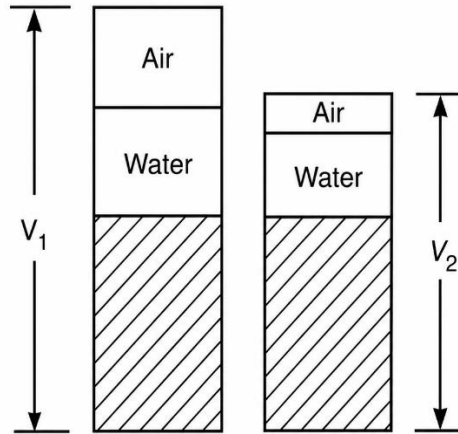


Section-A (25 Marks)

1. Define soil compaction and describe the effects of compaction on various soil properties. [1+4=5]

Answer:

Soil compaction is the process of densification of soil by removing the air voids present in the soil particles under the action of mechanical energy. This phenomenon changes the three-phase diagram of soil.

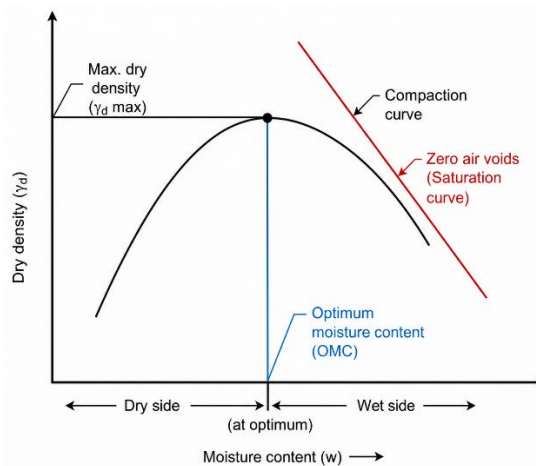


(a) Compaction

There are several effects of compaction on various soil properties. They are discussed in points below:

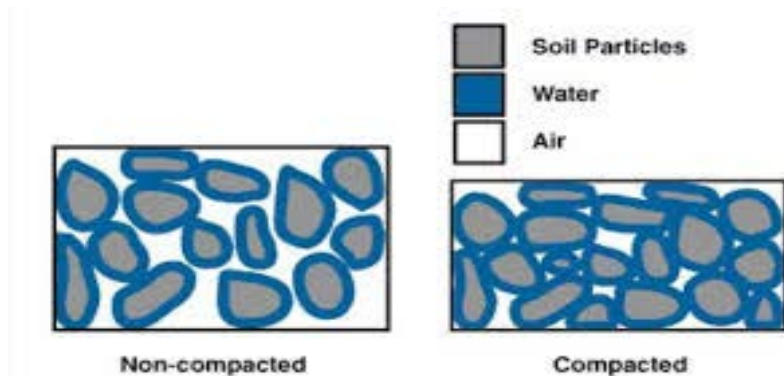
i. Dry Density:

Compaction process increases the dry density and unit weight of the soil mass. Maximum Dry Density (MDD) is achieved when the soil mass is compacted at the Optimum Moisture Content (OMC).



ii. Shear Strength of soil:

Compaction brings the soil particles together by expelling the air voids present in them and strengthens the interlocking which increases the shear strength of soil.



iii. **Bearing Capacity of soil:**

As the shear strength of the soil mass increases with the compaction, it ultimately increases the bearing capacity of the soil.

iv. **Permeability of soil:**

Compaction helps to reduce the loss of water in reservoirs, dams etc. as the voids are expelled out bringing the soil particles closer which reduces the permeability of soil.

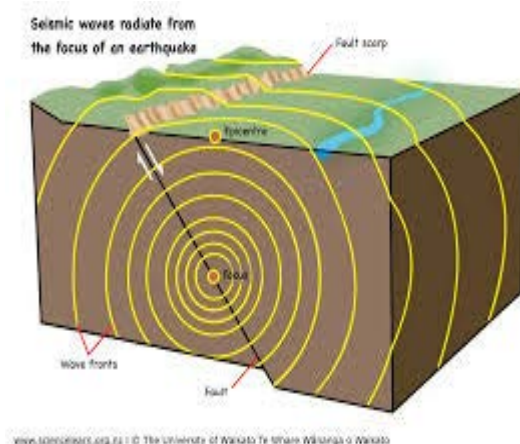
v. **Compressibility of soil:**

Compaction strengthens the shear strength and bearing capacity of soil mass which reduces the settlement of foundation, pavement failures and lateral deformation on slope.

2. **Describe the possible effects of an impending major earthquake disaster in Nepal. What precautions have been taken so far to manage the disaster? [5+5=10]**

Answer:

Earthquake is a natural calamity characterized by sudden, rapid shaking of the earth's surface either horizontally or vertically due to release of enormous energy stored in the earth's crust. Nepal is highly vulnerable to earthquakes. It ranks 11th among the countries with high vulnerability to earthquakes. Especially in the densely populated areas with unplanned urbanization, the effects of major earthquake are enormous costing lives of people and animals along with severe structural damage and collapse. Earthquake that struck Nepal with 7.8 Richter scale magnitude in 2072 Baisakh 12 caused huge damage and collapse of infrastructures, residential buildings as well as ancient heritages. Not only that, the disaster costed life of hundreds of people and livestock, affected the living conditions of many people, dropped the economy of the country and many more.



The possible effects of an impending major earthquake disaster in Nepal are described in points below:

i. **Damage and collapse of infrastructures and cultural heritages:**

The sudden shaking of earth surface in vertical and horizontal direction during earthquake, makes the aged cultural heritages and weak infrastructures more susceptible to severe damage and collapse. Similarly, unreinforced residential buildings made up of weak construction materials also has high risk of collapse during earthquake.

ii. Loss of life and injuries:

Earthquake also cost lives of people and livestock due to collapse and severe damage of the residential buildings, infrastructures and heritages. Major to minor injuries is also frequent in such disaster.

iii. Economic losses:

Earthquake directly or indirectly impacts the business of the earthquake affected areas. Reduction in tourism, collapse and damage to the industrial infrastructures, reduction in agricultural production due to earthquake causes huge economic losses.

iv. Landslides and other natural calamities:

Sudden movement of earth surface may also give rise to other natural calamities such as landslides and avalanche. The weak unstable slope might instigate huge landslide due to movement of earth. Similarly, the earthquake may also instigate movement on the mountains resulting in avalanche threatening settlements and mountain tourism.

v. Displacement and homeless:

Many people lose their home and get displaced to other areas due to collapse or severe damage of residential buildings by earthquake.

vi. Public health problems:

Shortage of food, water, medical services may result in the outbreak of epidemics due to unhygienic and poor sanitary conditions in the temporary camp.

The precautions that have been taken so far to manage the disaster are pointed out below:

i. Design of Earthquake Resistant Buildings:

Engineers and builders are encouraged for the design and construction of earthquake resistant buildings to prevent the severe damage and collapse of structures during major earthquakes. The main objective of earthquake Resistant Building is to minimize the case of deaths due to collapse of structures during earthquake.

ii. Building codes:

Nepal Building Code (NBC) has made a provision of NBC 105 for the seismic design of buildings. The buildings are only approved for construction if they are designed according to the prescribed codes by NBC.

iii. Disaster Risk Reduction and Management Act:

Government has also established several legal and institutionalized frameworks for the disaster management. National Disaster Risk Reduction and Management Authority (NDRRMA) coordinates and manages the disaster preparedness and response in Nepal.

iv. Retrofitting of buildings:

Public buildings such as schools, hospitals as well as some cultural heritages are strengthened via retrofitting to withstand earthquakes. However, seismic retrofitting is more preferred to temporarily strengthen the existing vulnerable important structures.

v. Emergency response planning:

Emergency operations centers have been established at national as well as local levels for preparation of impending natural disasters like earthquake. Ministry of Home Affairs has also officially identified and allocated open spaces in and around Kathmandu valley as safe assembly zones and temporary settlement sites.

3. What are the reasons for the landslides? What do you suggest to solve the problems? [4+6=10]

Answer:

Landslide is the movement of a mass of soil, rock or debris down the slope under gravity. It is caused when gravitational pull exceeds the strength of soil. In the context of Nepal, rugged topography, unstable hills with large unstable slopes are commonly found in mountain and hilly regions. These regions report frequent landslides especially in mid-hill regions, Chure and Mahabharat range. More than 80% of the territory is characterized by steep hills and mountains prone to landslides. Few examples of the occurrence of landslide are Jure landslide, Sindhupalchowk, Bhirkot village landslide, Dolakha, Mugling-Naraynghad highway landslide disasters and so on.



Some of the major reasons for the landslides are as follows:

i. Heavy rainfall:

Heavy rainfall causes seepage along the slope, increases pore water pressure which reduces the effective stress and shear strength of the soil. It also reduces friction between the soil particles due to which landslides occur.



ii. Seismicity:

Earthquakes and volcanic eruptions instigate landslides due to violent shaking of the ground surface. Seismic activities increase the shearing stress on the slope dislocating large volume of earth and rocks.

iii. Geological factors:

Geology comprising of weak and unstable slopes increases the risk of occurrence of landslides under the influence of earthquake or huge rainfall.

iv. Deforestation:

Trees and vegetative systems act as anchorage and reinforcement and holds the soil mass together. However, human intervention such as deforestation makes the soil mass loose and triggers landslides.

Some suggestions to solve the problems of landslide:

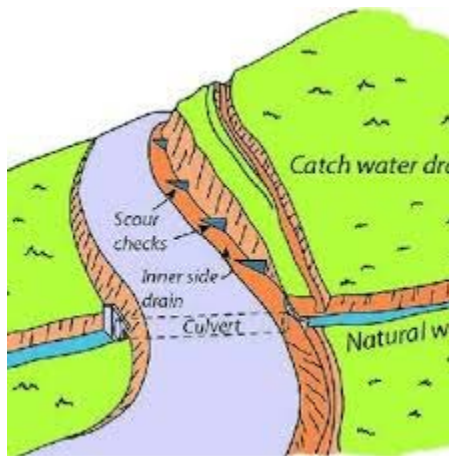
i. Afforestation and vegetation:

Forests and vegetative systems strengthen the soil mass and prevent soil erosion. Hence, focusing more on afforestation and vegetation helps to solve the problem of landslide.



ii. Proper drainage provision:

Landslides get triggered due to heavy and prolonged rainfall. Rainwater washes the soil particles and reduces the shear strength of the soil. Hence, proper drainage infrastructures such as cascade, catch drains, side drains must be provided to intercept the surface runoff and prevent frequent landslides triggered by surface runoff.



iii. Slope stabilization structures:

The frequency of landslides can be effectively minimized by providing slope stabilization structures such as retaining walls, gabion walls, geosynthetics, soil nailing etc.



iv. Planned land use:

Land use must be planned before human settlement and construction of infrastructures. These constructions must be avoided near landslide-prone zones.

v. Controlled construction activities:

Extreme care must be taken while carrying out construction activities. Sometimes, these construction activities itself trigger the unstable slope that results in occurrence of landslides.

Section-B (25 Marks)

4. Describe different methods of estimating the average precipitation over a basin. [5]

Answer:

Any form of solid or liquid water that falls from the atmosphere to the earth's surface is termed as precipitation. It can be rain, snow, drizzle, hail etc. However, most common form of precipitation is rainfall.

Average precipitation over a basin can be estimated by three methods. They are briefly described below:

i. Arithmetic average method:

If the rainfall values measured by rain gauge stations in catchment area varies less, then the average rainfall over an area can be calculated by average mean method.

Average rainfall over an area by Arithmetic average Method is given by:

$$P = \frac{P_1 + P_2 + P_3 + \dots + P_n}{n}$$

Where:

P = Average rainfall over the catchment (mm)

$P_1, P_2, P_3, \dots, P_n$ = Rainfall recorded at individual rain gauge stations (mm)

n = Number of rain gauge stations

ii. Thiessen Polygon Method:

- This method considers representative area for each rain gauge stations.
- Rainfall values of each rain gauge station are given weightage on the basis of the area closest to station.
- Average rainfall over an area by Thiessen Polygon Method is given by:

$$P = \frac{A_1 P_1 + A_2 P_2 + A_3 P_3 + \dots + A_n P_n}{A_1 + A_2 + A_3 + \dots + A_n}$$

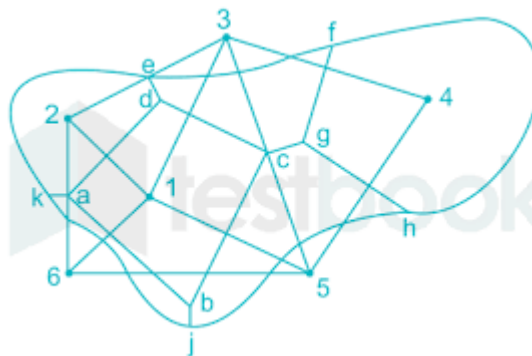
Where:

P = Average rainfall over the catchment area

$P_1, P_2, P_3, \dots, P_n$ = Rainfall at each rain gauge station

$A_1, A_2, A_3, \dots, A_n$ = Area of the Thiessen polygon corresponding to each station

n = Number of rain gauge stations



iii. Isohyetal method:

- Isohyetal method is one of the most accurate methods of determining average rainfall over an area.
- Consists of plotting of isohyets which is the line joining the points having same magnitude of rainfall.
- Area between the isohyets and catchment boundary is measured.
- The average rainfall over an area by the Isohyetal Method is calculated using:

$$P = \frac{\sum(A_i \times P_i)}{A}$$

Where:

P = Average rainfall over the catchment area

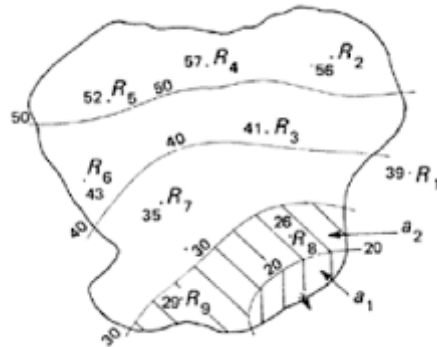
A_i = Area between two successive isohyets

P_i = Mean rainfall between two successive isohyets

$$P_i = \frac{P_1 + P_2}{2}$$

Where, **P₁** and **P₂** are the rainfall values of the two adjacent isohyets.

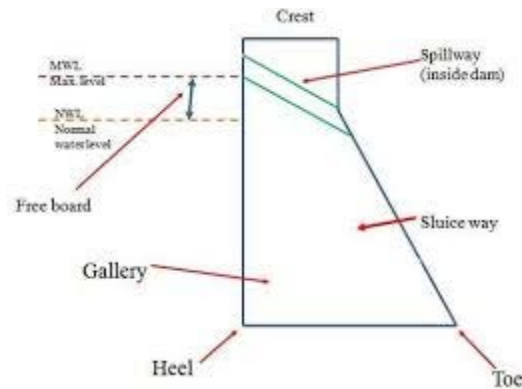
A = Total catchment area



5. How many types of failure occur in gravity dam? Describe their causes. [4+6 =10]

Answer:

Gravity dam is the permanent dam which resists all the forces acting on it by its own self weight. The forces acting on the gravity dam are weight of dam, horizontal and vertical component of water pressure, uplift pressure due to seepage, earthquake pressure and wind pressure.



There are four types of failure that can occur in gravity dam. They are described below along with their cause:

i. **Overturning failure:**

When the overturning moment due to water pressure and uplift pressure exceeds the resisting moment due to self-weight of the dam, then it rotates and topples about its toe.

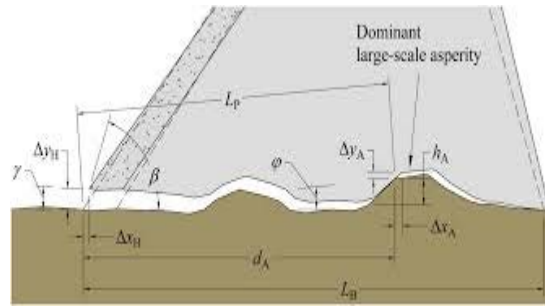
The **Factor of Safety (FOS) against overturning** is the ratio of the **stabilizing/resisting moment** to the **overturning moment** acting on the dam.

$$FOS_{\text{overturning}} = \frac{\sum M_R}{\sum M_O}$$

Where:

ΣM_r = Sum of resisting (stabilizing) moments about the toe of the dam

ΣM_o = Sum of overturning moments about the toe of the dam



ii. Sliding failure:

When the net horizontal forces like water pressure exceeds the shear and frictional resistance between the dam and foundation, then the dam begins to slide along the horizontal joint known as sliding failure of dam.

The **Factor of Safety (FOS) against sliding** is defined as the ratio of the total resisting forces to the total driving (sliding) forces.

$$FOS = \frac{\text{Resisting Forces}}{\text{Driving Forces}}$$

For a gravity dam:

$$FOS = \frac{\mu W + CA}{P}$$

Where:

μ = Coefficient of friction between dam and foundation

W = Effective weight of the dam (including vertical forces)

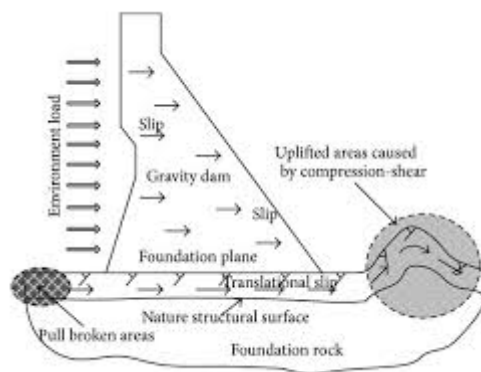
C = Cohesion between dam and foundation

A = Area of contact at the base

P = Total horizontal force causing sliding (mainly water pressure)

Simplified Formula (when cohesion is neglected)

$$FOS = \frac{\mu W}{P}$$



iii. Crushing failure:

If the compressive stress at heel or toe exceeds allowable crushing strength of masonry or concrete material. The maximum compressive stress generally occurs at the **toe** of the dam.

a. Direct Stress (σ_0)

$$\sigma_0 = \frac{W}{B}$$

Where:

W = Total weight of dam per meter length (kN)

B = Base width of dam (m)

b. Bending Stress Due to Eccentricity

$$\sigma_b = \frac{6We}{B^2}$$

Where:

e = Eccentricity of resultant force from the center of base (m)

c. Maximum Crushing Stress at Toe

$$\sigma_{max} = \frac{W}{B} \left(1 + \frac{6e}{B} \right)$$

$$\sigma_{max} = \sigma_0 + \sigma_b$$

d. Minimum Stress at Heel

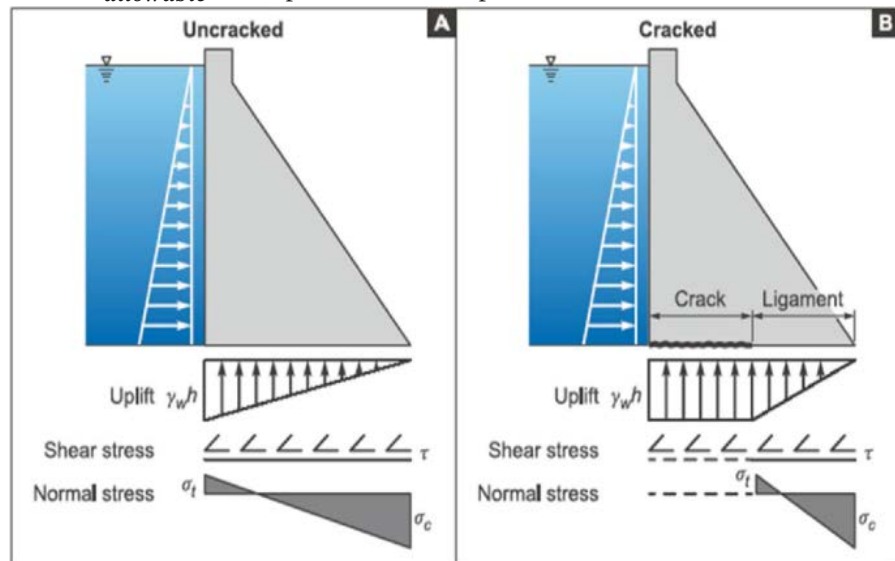
$$\sigma_{min} = \frac{W}{B} \left(1 - \frac{6e}{B} \right)$$

$$\sigma_{min} = \sigma_0 - \sigma_b$$

Crushing failure occurs when:

$$\sigma_{max} > \sigma_{allowable}$$

where $\sigma_{allowable}$ is the permissible compressive stress of the dam material.

**iv. Tension failure:**

Tensile failure of the dam occurs when the tensile stress developed in a dam section exceeds the tensile strength of the dam material. Since, both cement concrete and masonry dams are weak in tension, it is not allowed to be developed anywhere near the dam.

For a gravity dam, the stress distribution at the base is:

$$\sigma = \frac{W}{b} \pm \frac{6M}{b^2}$$

Where:

σ = Normal stress at the base (kN/m²)

W = Total vertical load (kN)

b = Width of the dam base (m)

M = Resultant moment about the center of the base (kN-m)

Maximum and Minimum Stress

$$\sigma_{max} = \frac{W}{b} \left(1 + \frac{6e}{b}\right)$$
$$\sigma_{min} = \frac{W}{b} \left(1 - \frac{6e}{b}\right)$$

Where:

e = Eccentricity of the resultant force from the center of the base

$$e = \frac{M}{W}$$

Condition for No Tensile Stress

For the dam to be safe against tensile failure:

$$\sigma_{min} \geq 0$$

Substituting:

$$\frac{W}{b} \left(1 - \frac{6e}{b}\right) \geq 0$$

Therefore,

$$e \leq \frac{b}{6}$$

This is known as the **Middle Third Rule**.

If $e > \frac{b}{6}$, then tensile stress develops at one edge of the dam base, leading to tensile failure.

Factor of Safety Against Tensile Failure

For gravity dams, tensile stress is generally not allowed. Hence the design requirement is:

$$e \leq \frac{b}{6}$$

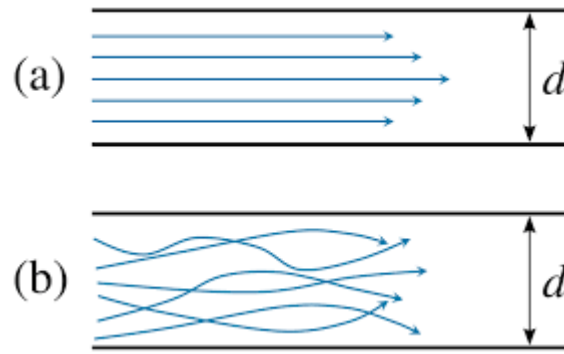
or $\sigma_{min} \geq 0$ to ensure safety against tensile failure.

6. What is the difference between laminar and turbulent flow? How citizen science can be used while measuring rainfall? Describe three applications of Bernoulli's equation. [2+2+6=10]

Answer:

There are two different types of fluid flow demonstrated by Osborne Reynold. The differences between laminar and turbulent flow are as follows:

Laminar Flow	Turbulent Flow
1. In laminar flow, fluid particles appear as a series of thin sheets sliding one over another.	1. Turbulent flow is characterized by intense mixing of fluid and random fluctuation in fluid velocity.
2. It is also called streamline flow or viscous flow.	2. It is also known as non-viscous flow or vortical flow.
3. Newton's law of viscosity is valid.	3. Newton's law of viscosity is not valid.
4. This flow occurs when $Re < 2000$	4. this flow occurs when $Re > 4000$



(a) Laminar flow

(b) Turbulent flow

Citizen science involves the participation of local communities and volunteers in collecting environmental data on a large scale. In rainfall measurement, citizens can act as observers by using low-cost rain gauges and mobile applications to record daily precipitation. Such practices have been widely adopted internationally. For example, the Community Collaborative Rain, Hail, and Snow Network (CoCoRaHS) in North America enables volunteers to submit daily precipitation data directly to meteorological agencies.

In a country like Nepal, with its rugged topography and many remote rural areas, citizen science offers a cost-effective way to collect essential environmental and meteorological data. The Smartphones For Water Nepal (S4W-Nepal) project has successfully engaged local citizens in data-scarce mountainous regions, including the Kathmandu Valley, to measure monsoon rainfall using simple PET-bottle rain gauges and Android smartphones. This approach has helped improve rainfall monitoring while reducing the need for expensive observation networks.

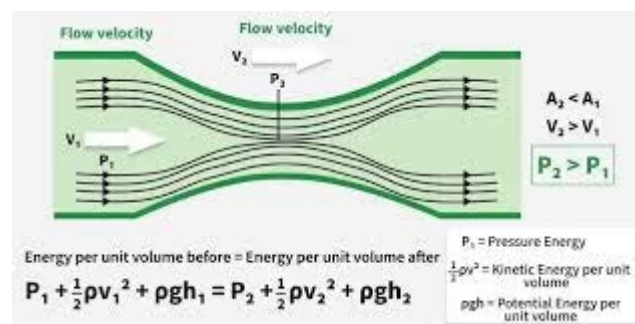
Bernoulli's principle is based on the principle of conservation of energy applied to the fluid. It states that for an ideal fluid flowing steadily through a pipe, the total energy remains constant along a streamline. The sum of pressure energy, kinetic energy, and potential energy per unit weight of fluid remains constant.

Bernoulli's Equation

$$\frac{P}{\gamma} + \frac{V^2}{2g} + z = \text{Constant}$$

or between two points,

$$\frac{P_1}{\gamma} + \frac{V_1^2}{2g} + z_1 = \frac{P_2}{\gamma} + \frac{V_2^2}{2g} + z_2$$



Where:

P = Pressure of fluid (N/m^2)

γ = Specific weight of fluid (N/m^3)

V = Velocity of fluid (m/s)

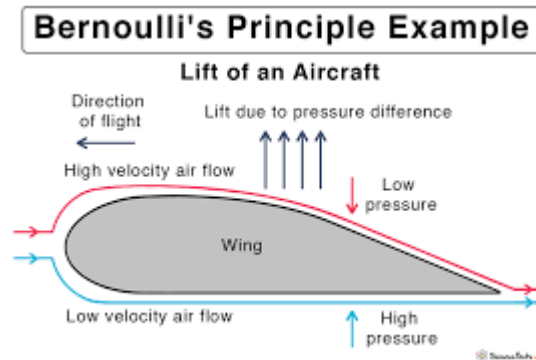
g = Acceleration due to gravity (9.81 m/s^2)

z = Elevation head above a datum (m)

Three applications of Bernoulli's principle:

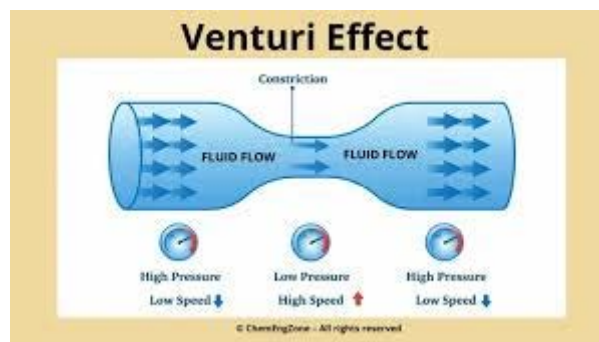
i. Aircraft wings:

Curved surface of aircraft wings forces air to move faster over top surface creating low pressure than the lower flat surface. This pressure difference creates the lift in the wings.



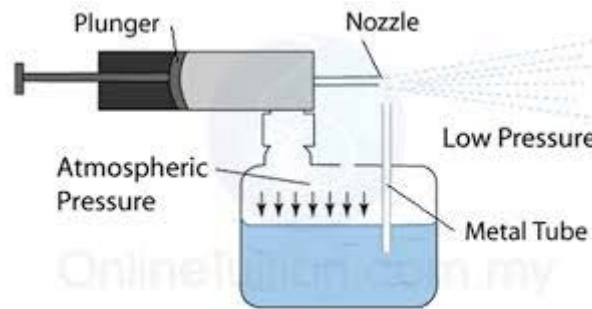
ii. Flow meters:

Devices such as orificemeter, venturimeter use Bernoulli's equation to determine fluid flow rates in pipe.



iii. Nozzle:

Air blown across the top of a tube moves at high speed so that the pressure above the tube decreases. The liquid is then pushed up the tube and sprayed as fine droplets.



Section-C (25 Marks)

7. List out the computational techniques and software used in design of highway civil structures. [5]

Answer:

Different computational techniques and software are used in design of highway civil structures.

Some of the computational techniques used for highway design are listed below:

- i. Finite Element Method:
This method is used for structural design and analysis of bridges, retaining walls, culvert etc.
- ii. Finite Difference Method:
It is used for solving differential equations in pavement and soil analysis.
- iii. Boundary Element Method:
This method is applied in solving geotechnical and foundation engineering problems.
- iv. Computer-Aided Design:
CAD is used for preparing highway alignments, plans, profiles, and cross-sections.
- v. Geographic Information system:

GIS is commonly used for route selection, mapping, transportation and water resources planning.

Some of the software used for highway design are listed below:

- i. **AutoCAD Civil 3D:**
This is a software used for the design of horizontal/vertical alignments, corridor assemblies, and earthworks.
- ii. **SAP2000:**
It is a finite element analysis software used for complex bridge design and dynamic loading.
- iii. **Plaxis 3D:**
It is a computational tool used for modeling of soil-structure interaction, design of tunnels, deep foundations and slope stability analysis.
- iv. **GeoStudio:**
It is a software used for the analysis of soil deformations, groundwater seepage and slope stability along highway cuts.
- v. **HEC-RAS:**
It is a standard computational technique used for modeling of water flow in natural rivers.
- vi. **ArcGIS/ QGIS:**
It is used to make maps, analyze data and environmental assessment.

8. What are the design and construction problems associated with hill roads? List the merits and demerits of river route and ridge route. [5+5=10]

Answer:

Hill road can be defined as the road that passes through the area with a cross slope of 25% or more. The hilly regions have extreme climatic condition, rugged topography and high altitude. The roads in these areas are affected by landslides, soil erosion and other problems.

Some of the major design and construction problems associated with hill roads are as follows:

- i. **Topographical challenges:**
Highly broken relief, steep slope, deep gorges and large number of watercourses may unnecessarily increase the road length.
- ii. **Geological and climatic variations:**
Great variation in the geological, hydrological and climatic conditions from one place to another.
- iii. **Slope instability:**
Unstable slope after road construction triggers landslides, rockfalls.
- iv. **Installation of various road structures:**
Hill roads require construction of various road structures such as retaining walls, parapet walls, breast walls, side drains, catch-water drains and others.
- v. **Design of hairpin bends to gain height:**
Steep elevated area requires hairpin bends and sharp curves which restricts the vehicle speed and other geometric design parameters of hill road.

Hill road is characterized by longer route with sharp curves. There are two possible route location viz (i) river route and ridge route.

i. River route:

River route is the route located along the river valley and is the most frequent case of hill road alignment.



Merits:

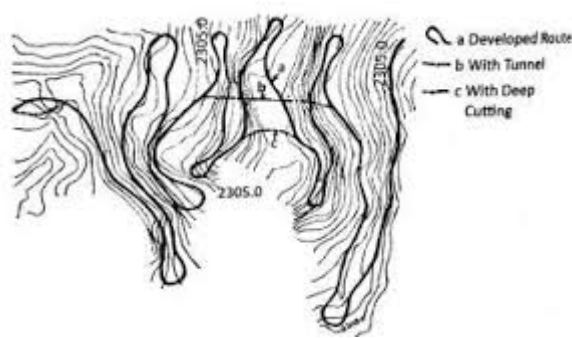
- River route has comparatively gentle gradient and serves rural settlement.
- Availability of water and other construction materials along the river route.
- River route incorporates low vehicle operating cost.

Demerits:

- Numerous horizontal curves must be provided along the river route.
- Extensive earthwork is required in this route.
- Special retaining and protection of walls on the hill side must be provided in river route.
- Massive river training and protection works must be provided on the river side.

ii. Ridge route:

Ridge route is the route located along the ridge of hills.

**Merits:**

- Ridge route is selected on comparatively stable geology and mild slope conditions.
- It helps to avoid water crossings and reduce the number of cross-drainage structures along the route.

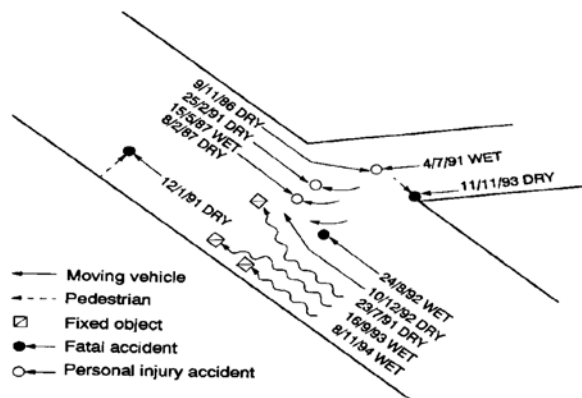
Demerits:

- Ridge route possesses very steep gradient.
- It consists of large number of sharp curves including hairpin bends.
- Construction of special structures are required along the ridge route.

9. Explain the causes of road accidents in urban areas and suggest suitable traffic engineering and policy measures to reduce them. [4+6=10]

Answer:

Urban roads are characterized by the mixed traffic flow with high density and high pedestrian volumes which increases the risk of road accidents. In Nepal, nearly half of all the traffic accidents throughout the country occur in Kathmandu valley alone. It was estimated about 8000 fatalities in Nepal due to road crashes.



There are several causes of road accidents. Some of them are explained in points.

i. Driver:

Excessive speeding, recklessness, violation of rules and regulations, failure to understand traffic signals, temporary effects due to fatigue, sleep or alcohol, inexperience of driver etc. causes road accidents.

ii. Pedestrian:

- Violation of rules and regulations by pedestrians, use of carriageway meant for vehicular use etc. causes road accidents.
- iii. Road conditions:
Skidding road surface, potholes, ruts and corrugation, inadequate road markings etc. are the causes of road accidents.
 - iv. Road design:
Defective geometric design such as inadequate distance, inadequate width and shoulder, improper lighting, improper traffic control devices cause road accidents.
 - v. Weather:
Unfavorable weather conditions such as fog, snow, dust, smoke etc. restricts normal visibility which makes the driving unsafe.

Preventive measures for road accidents are classified into three categories:

- i. Engineering measures:
 - Geometric elements of the road (sight distance, width of pavement, horizontal and vertical alignment) are checked and corrected if necessary.
 - Periodic maintenance of brakes, steering, lighting system of vehicles should be carried out.
 - Accident in night time can also be decreased by providing proper road lighting.
 - Pavement surface characteristics (including skid resistance values) must be checked and suitable maintenance measures are taken to maintain the road up to design standards.
 - Necessary construction of by-passes to separate through traffic from local traffic.
 - Design and construction of grade separated intersections or flyovers to minimize delay and conflicts.
- ii. Enforcement measures:
 - Speed control rule enforcement to develop correct speed habits. Surprise check on spot speed of vehicles at different locations and taking legal actions on those who violates the speed limit.
 - Strict test while issuing driving license, testing of driving skill while renewing.
 - Heavy penalties levied for defective vehicles.
 - Observation and modification of existing rules and regulations.
 - Proper vision checks and reaction time checks at prescribed intervals.
- iii. Education measures:
 - It is very important to educate road users for the precaution measures to use road facilities with safety.
 - Passengers, pedestrians, drivers must be taught the rule of road, correct manner of crossing.
 - Proper driving training works must be made available.

Section-D (25 Marks)

10. What is Environment Impact Assessment (EIA)? Describe its importance in project development. [2+3=5]

Answer:

- Environment Impact Assessment (EIA) is a major tool that checks whether the economic development project is environmentally sustainable or not.
- **According to Nepal's Environment Protection Act, 2076 (2019)**, an Environmental Impact Assessment (EIA) is defined as a detailed study or evaluation to determine if a proposal causes adverse environmental impacts during implementation. The process includes mandatory measures to avoid or mitigate such impacts, with procedures for scoping, reporting, and approval governed by the Environment Protection Rules.
- EIA was first used in USA in 1970 and it became widely used throughout the world after UN Earth Summit 1972.
- In Nepal, EIA is mandatory for large scale projects such as:

- a. **Hydropower projects with the installed capacity greater than 50 MW** such as Upper Tamakoshi Hydropower Project, Tallo Seti Hydropower Project
- b. **highways having length greater than 50 km** such as Narayanghat-Butwal Road project, Kathmandu Terai fast-track
- c. **irrigation project having command area greater than 2000 hectares:** for example, Nagmati dam project, Sikta irrigation project
- d. **airports** such as Gautam Budhha International Airport, Nijgad International Airport.
- e. **railways** such as east-west electric railway and so on.

➤ Basis for EIA are: EPA 2076 and EPR 2077.

Importance of EIA in project development:

- i. Risk reduction and management:
 - EIA assesses the impacts of development project on the environment and natural resources.
 - It helps to reduce environmental impacts of developmental projects.
 - It identifies the environmental risk of project on ecologically fragile landscapes.
- ii. Reduction of overall project cost:
 - EIA also helps to reduce overall economic and environment cost of project.
 - It also increases project benefits.
 - EIA also prevents the loss of investment on the infrastructures that are ecologically and environmentally unviable.
- iii. Sustainable development:
 - EIA encourages sustainable development, which supports the construction of environmentally feasible infrastructures.
 - EIA believes that development and environment conservation can go hand in hand when it is planned, studied and analyzed thoroughly.
- iv. Informed decision making and community trust:
 - Government bodies, investors, planners thoroughly investigate the project and decide whether to implement or disapprove the project on the basis of all the collected information.
 - EIA also encourages compulsory public hearing to understand the sentiments of the people residing on project affected area.

11. Classify the water supply systems. List and describe the steps involved in pipe-laying for a water supply network. [4+6=10]

Answer:

Water supply system is the system which ensures the supply of safe, potable water to the consumer's household in adequate quality and quantity. Water supply systems are classified based on the method of distribution and layout of the distribution network.

- i. Based on method of distribution:
 - a. Gravity system:

It is the system of water supply in which the source of water is located at the elevated area than that of distribution area which allows the water to flow through transmission mains under gravity.
 - b. Pumping system:

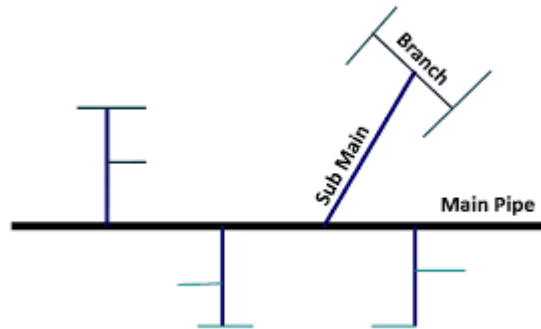
It is the system of water supply in which the source of water is located below the distribution area so that water has to be pumped directly to distribution mains without providing elevated storage.
 - c. Combined gravity and pumping system:

It possesses the characteristics of both gravity and pumping water supply systems in which water is pumped into distribution mains and excess water is then stored to elevated reservoir which supports gravity flow.

ii. Based on layout of distribution network:

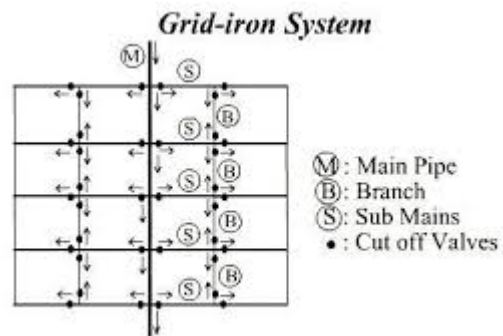
a. Dead-end system / Tree system:

One main pipeline is laid at the center of the area to be served. From the both sides of main pipeline, submains take off which is further divided into branches and then the service connections are provided.



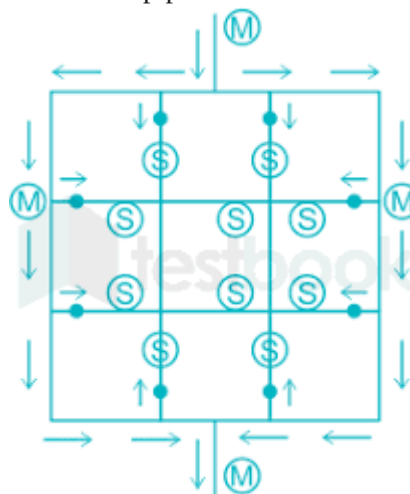
b. Grid-iron system:

One main pipeline is laid through the center of area to be served, from both side of mains, submains take off in perpendicular directions. In this system, main, submain and branches are interconnected with each other.



c. Ring system / Circular system:

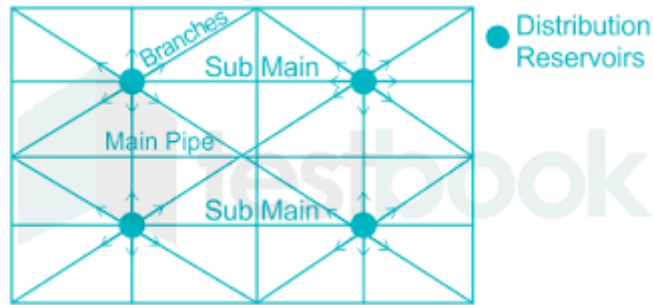
This system is suitable in well planned cities where road networks are planned. One main pipeline is laid on the periphery of the area to be served in rectangular or circular blocks. Then submains take off from the main pipelines towards inner area.



d. Radial system:

This system is the reverse of circular system in which water flows towards outer periphery from it. In radial system, the area to be served is divided into several zones which is provided with

distribution reservoirs, from where water is supplied radially to different connections towards outer periphery.



Steps involved in pipe laying for water supply network:

- i. Preparation of detailed map:**
Survey is conducted for the preparation of detailed map with the alignment of pipelines from source to treatment plant and distribution areas including existing sewer pipelines and position of road.
- ii. Locating proposed alignment on the ground:**
Proposed alignment of pipelines on the ground is located by driving central stakes 30 m apart for straight reaches, and 7.5-15 m apart for curved road.
- iii. Location of pipes:**
Pipes can be laid on ground surface or below the ground surface. If the pipes are to be laid underground, trenches must be excavated.
- iv. Excavation and preparation of the trench:**
Excavation of trench can be done mechanically or manually according to the specification. Trench should be excavated 30 to 45 cm more than the external diameter of the pipe. More than 90 cm clearance is provided above the crown. Bottom of the trench must be carefully prepared to prevent the settlement when the pipes are laid off.
- v. Dewatering of trench:**
If ground water table is high then water from the trench must be dewatered.
- vi. Lowering the pipe on trench:**
Pipelines are lowered on the trench carefully without any damage to the pipe.
- vii. Jointing of pipes:**
Pipes are jointed to each other by the help of suitable joints.
- viii. Testing of pipes:**
After laying and jointing of the pipe, test is carried out at the section of 500 m length. One end of the pipe is closed and from other end water is pumped at double working pressure for 24 hours. The leakage from the pipe must be within permissible limits.
- ix. Disinfection and backfilling:**
Disinfection can be done by adding chlorine for 12 hours. Backfilling of trench is done by parent material and compaction up to 15 cm from ground level.



12. Compare the separate and combined sewer systems. Define self-cleaning velocity and non-scouring velocity in sewers. Why sewers are not designed for full-flow conditions? Explain in short. [4+3+3=10]

Answer:

Separate and combined sewerage systems are the two common types of sewerage systems. The comparison between them is listed in points below:

Separate sewer system	Combined sewer system
1. This sewerage system consists of two different sewers, one for convergence of sewage and another for rainwater.	1. This sewerage system consists of only one sewer to carry both sanitary sewage as well as stormwater.
2. It requires sewer of large area since cleaning is not possible in small sewers.	2. It requires large dimensions of sewer.
3. Load on treatment plant is reduced as waste type is collected separately.	3. Load on treatment plant increases as waste type is collected at same sewer.
4. The system reduces possibility of pollution due to overflow by excess rainfall.	4. Possibility of overflow of sewage during rainy season.
5. Choking problem can be noticed in separate sewer system as self-cleansing velocity is difficult to be achieved.	5. Self-cleansing velocity is easily achieved in combined sewer system.

Self- cleansing velocity:

- It is the minimum velocity at which the solid particles remains in suspension without settling at sewer's bottom.
- It is the velocity at which even scour of the deposited particles of the given size will take place.
- It must be maintained at least once a day as it is not possible to maintain the velocity throughout the day.

Formula for Self-Cleansing Velocity

According to Shield's Theory:

$$V_s = \sqrt{\frac{8K(\rho_s - \rho)gd}{f\rho}}$$

Where:

V_s = Self-cleansing velocity (m/s)

K = Dimensionless constant

ρ_s = Density of solid particles (kg/m³)

ρ = Density of water (kg/m³)

g = Acceleration due to gravity (9.81 m/s²)

d = Diameter of particle (m)

f = Friction factor

Non-scouring velocity in the sewer:

- It is the maximum velocity at which interior sewer surface is not damaged.
- It is the velocity at which no scour or abrasion takes place.
- Non-Scouring Velocity prevents erosion of channel bed and banks and ensures stability of sewers.
- **Formula for Non-Scouring Velocity**

According to Kennedy's Theory:

$$V_0 = 0.55 m D^{0.64}$$

Where:

V_0 = Non-scouring (critical) velocity (m/s)

m = Critical velocity ratio

D = Depth of flow (m)

For average silt conditions:

$$m = 1.0$$

Therefore,

$$V_0 = 0.55 D^{0.64}$$

Sewers are not designed for full flow conditions because:

i. Variation of sewage flow:

Sewage flow varies throughout the day. Hence, to cater the peak sewage flow during storms, infiltration and future population growth, sewer is designed for the partial flow conditions.

ii. Ventilation:

The space left in partially flowing condition in sewer allows air circulation which prevents accumulation of harmful and foul gases such as Hydrogen Sulphide gas, methane gas etc.

iii. Prevention of overflow:

If sewer runs in full flow condition, slight increase in discharge causes overflowing which results in unhygienic condition in the area.

